

Problem Set 2: Selection on Observables

MY457 | WT 2026 | Week 4

1 Concepts

- 1.1 A researcher wants to estimate the effect of electoral gender quotas on women's representation. Their unit of analysis is parties. Their treatment variable is a binary indicator for whether a party adopted a gender quota or not. They decide to condition on party ideology (left, center, right). Assuming that the conditional independence assumption holds, what other issue might they face when using selection on observables to estimate the treatment effect? A stylised version of their data is shown in Table 1:

Table 1: Stylised Dataset

Party	Ideology (X)	Quota (D)	Share of female Candidates (Y)
A	Left	1	0.45
B	Center	1	0.30
C	Right	0	0.20
D	Left	1	0.55
E	Center	0	0.28
F	Right	0	0.50

Solution

The researcher might face the issue of common support (or overlap). This means that for each value of the covariates (in this case, party ideology), there should be both treated and control units. If, as in this example, all left-wing parties adopted gender quotas while all right-wing parties did not, there would be no common support for these two groups. This lack of overlap would make it impossible to estimate the treatment effect for those ideologies, as there would be no comparable control group for treated units (or vice versa).

1.2 Which estimand(s) could the researchers identify?

 Solution

Even if the common support assumption is violated, the researchers could still identify Conditional Average Treatment Effects for those party ideologies for which there is variation in treatment status. For example, if some centrist parties adopted gender quotas while others did not, the researchers could estimate the ATT for centrist parties.

1.3 Explain the notation $(Y_1, Y_0) \perp D|X$. Does this mean that $(Y_1, Y_0) \perp X$?

 Solution

$(Y_1, Y_0) \perp D|X$ represents the conditional ignorability assumption. It states that, conditional on X , potential outcomes are independent of treatment status. In other words, if X is held constant (e.g. if we only compare units with the same value of X), the potential outcomes and the treatment are unrelated.

Importantly, this doesn't mean that the potential outcomes are independent of X (which is what the expression $(Y_1, Y_0) \perp X$ states). There may well be a causal relationship between X and the potential outcomes. In fact, if X is a candidate confounder, we would expect this!

1.4 Consider the causal graph below. List all paths between D and Y .

```
dag1 <- dagify(  
  Y ~ D + X + M + P,  
  D ~ X + Q,  
  M ~ D,  
  X ~ Q,  
  R ~ D,  
  R ~ Y,  
  P ~ Q,  
  exposure = "D",  
  outcome = "Y",  
  coords = data.frame(  
    name = c("D", "Y", "X", "M", "Q", "P", "R"),  
    x = c(0, 4, 1, 2, 3, 4, 2),  
    y = c(0, 0, 2, -2, 2, 2, -1)  
  )  
)
```

```
)
ggdag1 <- ggdag(dag1) +
  theme_dag()
ggdag1
```

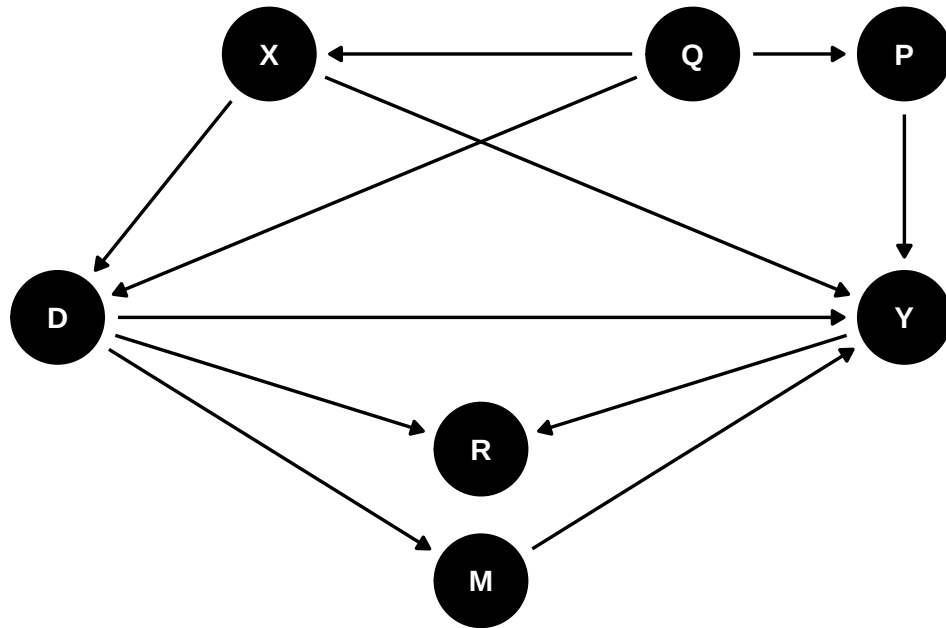


Figure 1: Causal Graph

💡 Solution

The paths between D and Y are:

1. $D \rightarrow Y$
2. $D \leftarrow X \rightarrow Y$
3. $D \leftarrow Q \rightarrow X \rightarrow Y$
4. $D \leftarrow Q \rightarrow P \rightarrow Y$
5. $D \rightarrow M \rightarrow Y$
6. $D \rightarrow R \leftarrow Y$

- 1.5 Define a conditioning set that is sufficient to identify the effect of D on Y . Explain your answer.

Solution

To identify the effect of D on Y , we need to specify a conditioning set that blocks all backdoor paths between D and Y and does not include any descendants of D .

One possible conditioning set that blocks all backdoor paths between D and Y is $\{X, P\}$. Conditioning on X blocks paths 2 and 3, while conditioning on P blocks path 4. Path 5 is not a backdoor path because it doesn't start with an arrow into D . Hence, we don't include M in the conditioning set. Path 6 is blocked because R is a collider (a collision node) that is outside the conditioning set and R has no descendant in the conditioning set (its only descendant is Y).

Alternatively, we could also condition on $\{Q, X\}$, which would block path 2 through X and paths 3 and 4 through Q . Again, we do not include M or R in the conditioning set for the same reasons as above.

2 Simulation

- 2.1 Suppose the government wants to implement an intensive course D , designed for recently unemployed people to help them get back into the job market and improve their salaries (Y). As the resident data scientist you argued for D to be randomized, but unfortunately you were overruled by your seniors who raised ethical concerns. They do, however, let you measure two covariates, $X1$ and $X2$. Below we simulate a dataset that contains all relevant variables. This also includes two unobserved variables $U1$ and $U2$ that naturally wouldn't be contained in your dataset were this real data. Draw a DAG that represents the data generating process. Helpful packages for drawing and visualising DAGs include `dagitty` and `ggdag`.

```
set.seed(123)
n_obs <- 10000
tau <- 25000

# Create the dataset
sim_data <- tibble(
  U1 = rbinom(n_obs, 1, 0.5),
  U2 = rbinom(n_obs, 1, 0.5),
  X2 = sample(1:50, n_obs, replace = TRUE),
  X1 = 10 + 1500 * U1 + 1500 * U2 + rnorm(n_obs, mean = 0, sd = 100),
  Y0 = 20000 + 1000 * X2 + 1500 * U2 + rnorm(n_obs, mean = 0, sd = 1000),
```

```

Y1 = Y0 + tau,
prob_d = pmin(pnorm(X2, mean = 50, sd = 25) + 0.5 * U1, 1),
D = rbinom(n_obs, 1, prob_d),
Y = ifelse(D == 1, Y1, Y0)
)

```

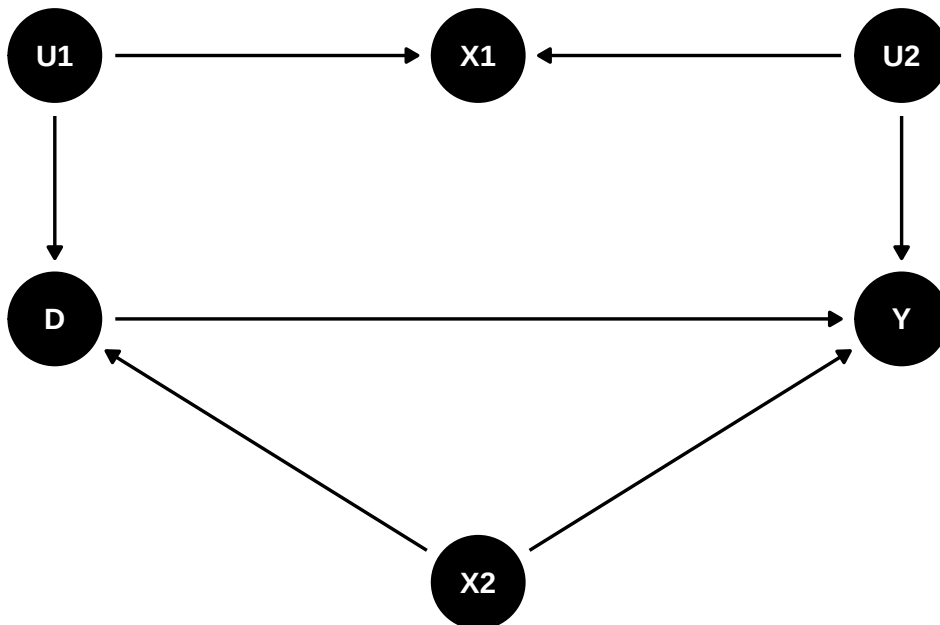
💡 Solution

```

dag2 <- dagify(
  X1 ~ U1 + U2,
  Y ~ U2 + X2 + D,
  D ~ U1 + X2,
  exposure = "D",
  outcome = "Y",
  coords = data.frame(
    name = c("D", "Y", "X1", "X2", "U1", "U2"),
    x = c(0, 2, 1, 1, 0, 2),
    y = c(0, 0, 1, -1, 1, 1)
  )
)

ggdag(dag2) +
  theme_dag()

```



2.2 Based on your DAG, which variables would you need to control for to estimate the effect of D on Y ? Briefly justify your answer.

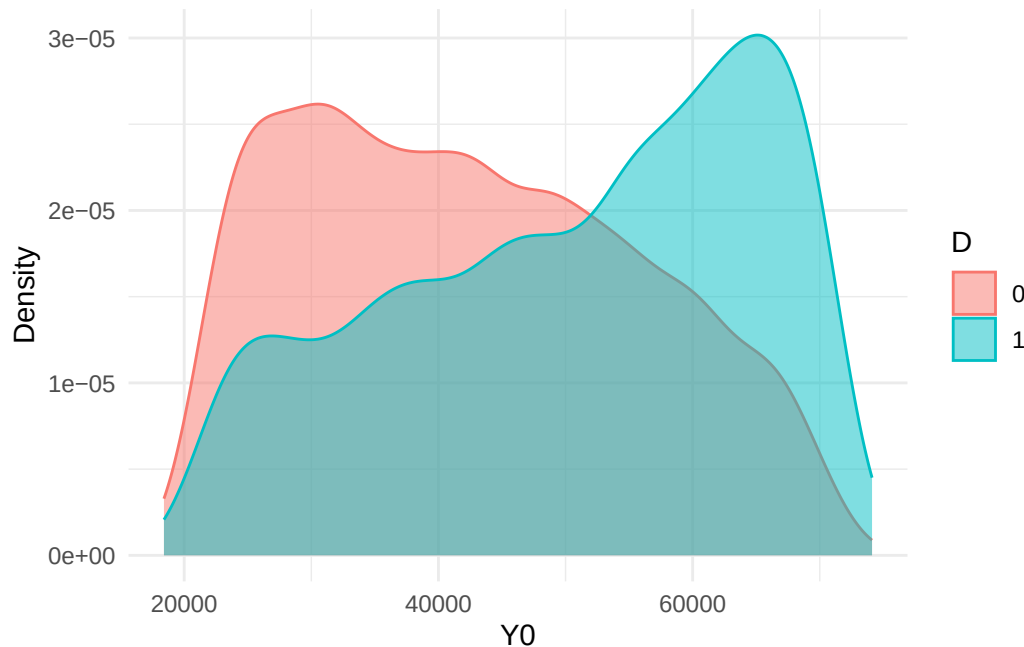
💡 Solution

To estimate the effect of D on Y , we need to control for $X2$. This is because $X2$ is a confounder that affects both the treatment assignment (D) and the outcome (Y). By controlling for $X2$, we can block the backdoor path from D to Y through $X2$, thus isolating the causal effect of D on Y . We shouldn't control for $X1$ because it is a collision node on the path $D \leftarrow U1 \rightarrow X1 \leftarrow U2 \rightarrow Y$. Because $X1$ is a collision node, it blocks the path $D \rightarrow U1 \rightarrow X1 \leftarrow U2 \rightarrow Y$ which implies that we don't have to include $U1$ and $U2$ in the conditioning set. As they represent unobserved variables, we wouldn't be able to control for them in practice anyway.

2.3 Generate a clear and compelling visualisation that shows the distribution of just Y_0 , split by treatment status. Do you think we can assume that $(Y_1, Y_0) \perp D$?

💡 Solution

```
ggplot(sim_data) +  
  aes(x = Y0, fill = factor(D), color = factor(D)) +  
  geom_density(alpha = 0.5) +  
  xlab("Y0") +  
  ylab("Density") +  
  labs(fill = "D", color = "D") +  
  theme_minimal()
```



The graph shows the distribution of Y_0 for both treatment groups ($D = 0$ and $D = 1$). The two distributions appear to be different, with the control group ($D = 0$) having a higher density of lower Y_0 values compared to the treatment group ($D = 1$). This suggests that the potential outcomes are not independent of treatment status, i.e., $(Y_1, Y_0) \not\perp D$.

2.4 Check for balance in the covariates $X1$ and $X2$. What do you find?

💡 Solution

```
lapply(c("X1", "X2"), function(x) {
  t.test(sim_data[[x]] ~ sim_data$D) |>
    tidy() |>
    mutate(variable = x)
}) |>
  bind_rows() |>
  select(variable,
    DiM = estimate,
    Mean_D0 = estimate1,
    Mean_D1 = estimate2, p.value
  ) |>
  kable()
```

variable	DiM	Mean_D0	Mean_D1	p.value
X1	-769.745123	1143.75361	1913.49873	0
X2	-8.395215	21.63144	30.02665	0

The t-tests for both $X1$ and $X2$ show significant differences in means between the treatment and control groups.

- 2.5 Imagine you only had access to the observed data and didn't know anything about the data generating process. Based on the results of the balance tests, would you feel confident to identify a conditioning set that would allow you to estimate the ATE without bias? Why or why not?

Solution

No, based on the balance tests alone, we would not be able to identify a conditioning set that would allow us to estimate the ATE without bias. Both $X1$ and $X2$ show significant differences between the treatment and control groups, but this alone tells us nothing about whether they are confounders or not. In fact, $X1$ is not a confounder but a collider, so conditioning on it would actually introduce bias. More generally, any variable that is associated with the treatment — whether because it shares a common cause with the treatment, is a collider, or is caused by the treatment — may be imbalanced. Imbalance alone does not tell us whether conditioning on a variable will remove or introduce bias.

- 2.6 As the resident data scientist you are faced with the task of estimating the effect of the course (D) on observed salaries (Y). First, naïvely estimate the ATE using a linear regression and display the results in a regression table showing the coefficients and 99.9% confidence intervals. How well does the estimated ATE match the true ATE?

💡 Solution

```
models <- list()
models$naive <- lm_robust(Y ~ D, data = sim_data)

modelsummary(models,
  gof_map = c("nobs", "r.squared"),
  stars = T,
  fmt = 0,
  statistic = "conf.int",
  conf_level = 0.999,
  notes = "99.9% confidence intervals in brackets",
  output = "markdown"
)
```

	naive
(Intercept)	42363*** [41771, 42956]
D	33406*** [32484, 34327]
Num.Obs.	10000
R2	0.591
• $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$	

Our true ATE is set to be 25,000. Our estimate is 33,405, almost 10,000 above the true one. The 99.9% confidence interval [32,484; 34,327] does not cover the true ATE, providing strong evidence of selection bias. In fact, even with very wide confidence intervals (99.9%), we reject the null hypothesis that the true effect equals 25,000. Our estimator yielded a very precise but severely biased estimate of the ATE.

2.7 Estimate two more linear regressions: one that includes X_2 as a control and one that includes X_2 and X_1 as controls. Display all three regression results in a single table showing the coefficients and 99.9% confidence intervals. Which estimator yields an unbiased estimate of the estimand (ATE)? Which estimator(s) are biased?

Solution

```
models$condition_on_X2 <- lm_robust(Y ~ D + X2, data = sim_data)
models$condition_on_X1_X2 <- lm_robust(Y ~ D + X2 + X1, data = sim_data)

modelsummary(models,
  gof_map = c("nobs", "r.squared"),
  stars = T,
  fmt = 0,
  statistic = "conf.int",
  conf_level = 0.999,
  notes = "99.9% confidence intervals in brackets",
  output = "markdown"
)
```

	naive	condition_on_X2	condition_on_X1_X2
(Intercept)	42363*** [41771, 42956]	20718*** [20633, 20803]	19960*** [19872, 20049]
D	33406*** [32484, 34327]	25005*** [24918, 25092]	24524*** [24440, 24608]
X2		1001*** [998, 1004]	1005*** [1003, 1008]
X1			1*** [1, 1]
Num.Obs.	10000	10000	10000
R2	0.591	0.997	0.997

- $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

The estimator that includes X_2 as a control yields an estimate very close to the true ATE, with an estimate of 25,005 and a 99.9% confidence interval of [24,086; 25,924] that covers the true ATE of 25,000. Adding X_1 as a control induces bias because X_1 is a collider on the path $D \leftarrow U_1 \rightarrow X_1 \leftarrow U_2 \rightarrow Y$. By conditioning on X_1 , we open this path and create a spurious association between D and Y through U_1 and U_2 .

3 Replication

In this section, we will use real-world data to reinforce what we have learned. We will analyse the dataset employed in *Dollars on the Sidewalk: Should U.S. Presidential Candidates Advertise in Uncontested States?*.

In the USA, political parties spend over 750 million dollars on television adverts during campaigns. However, because of the nature of the US political system, many parties only advertise in states they deem competitive. Yet, beyond votes, advertising might also increase the amount of financial contributions people make to political campaigns. The authors study whether ads lead to greater levels of contributions to campaigns. To do this, they consider areas in non-competitive states that are exposed to ‘spillover’ ads from bordering competitive states as treated units, and areas in non-competitive states that are not exposed to these ads as control units.

- 3.1 Skim the paper and fill out the table below to provide a summary of the paper’s research design. Be as precise and concise as possible. Edit the table directly in your .qmd to provide your answers.

Category	Description
Main Hypothesis	
Unit of Analysis	
Treatment	
Outcome	
Identification Strategy	
Estimand	

Solution

Category	Description
Main Hypothesis	Exposure to political ads increases campaign contributions
Unit of Analysis	Zip codes
Treatment	Exposure to at least 1000 ads
Outcome	Campaign contributions
Identification Strategy	Propensity score matching to estimate the ATT of ad exposure on campaign contributions
Estimand	Average Treatment Effect on the Treated (ATT)

3.2 Why can't the authors just compare campaign contributions between zip codes in non-competitive states that are exposed to spillover ads and those that are not?

 Solution

The authors can't just compare campaign contributions between the two groups of zip codes because there may be confounding factors that affect both the likelihood of being exposed to spillover ads and the level of campaign contributions. For example, zip codes that are closer to competitive states may have different demographic characteristics, economic conditions, or political engagement levels compared to those that are farther away. These differences could influence both the exposure to ads and the amount of contributions, leading to biased estimates of the treatment effect if not properly accounted for.

3.3 Which variables are in the conditioning set that the authors use to estimate the propensity score? Is there a variable you would have added to the conditioning set that the authors didn't include? Justify your answer.

 Solution

The authors use the following variables in their conditioning set to estimate the propensity score: Median Household Income, Percentage of Hispanics, Percentage African American, Percent College Graduates, Population Density, and State fixed effects.

All variables that causally affect both the treatment assignment (exposure to ads) and the outcome (campaign contributions) should be included in the conditioning set. Leaving the conditioning set $\mathbf{X}$ unchanged implies that conditioning on the above variables is sufficient to block all backdoor paths between the treatment and the outcome and that $(Y_{1i}, Y_{0i}) \perp D_i \mid \mathbf{X}_i$ holds.

We already condition on median household income, but we could also include a measure of wealth, such as the average net worth of households in the postcode. If we believe that wealth has a causal effect on being exposed to ads and on making a campaign contribution, it would be an additional confounder.

- 3.4 Read into R the replication data file `dollars_on_the_sidewalk.dta` and assign it to an object called `data`. These data are at the zip code level, and include data from both competitive states and non-competitive states, indicated by the dummy variable `NonComp` which takes a value of 1 if non-competitive. For only those zip codes in non-competitive states, define a binary treatment variable (`D`) based on whether the zip code received more than 1,000 ads (`TotAds`). Remove rows that contain missing values on any of the variables in the dataset. How many units are in the treated group, and how many are in the control group?

 Solution

```
data <- read_dta(here("2026", "soo", "data", "dollars_on_the_sidewalk.dta"))

data <- data |>
  filter(NonComp == 1) |>
  mutate(D = ifelse(TotAds > 1000, 1, 0)) |>
  na.omit()

data |>
  count(D) |>
  kable()
```

D	n
0	11165
1	6237

We have 11,165 zip codes in control and 6,237 in treatment.

- 3.5 What is the mean campaign contribution (`Cont`) for each level of `D`? Estimate the naïve difference-in-means using these two quantities.

 Solution

The difference-in-means is 0.24.

```
data |>
  group_by(D) |>
  summarise(
    Cont = mean(Cont, na.rm = T)
  ) |>
  kable()
```

D	Cont
0	19.91722
1	20.16180

```
dim <- mean(data$Cont[data$D == 1]) - mean(data$Cont[data$D == 0])
```

- 3.6 The authors use propensity score matching to estimate the ATT. Using the authors' conditioning set, fit a logistic regression to estimate the propensity score for each unit in the data. (Hint: Use `family = binomial(link = "logit")` in the `glm` function. To predict probabilities, use `predict()` but remember to set `type = "response"`.) Note that the authors include the following covariates in their conditioning set: `Inc`, `PercentHispanic`, `PercentBlack`, `per_collegegrads`, `density`, `State`. Add the propensity scores to your dataset as a new variable called `p_score`.

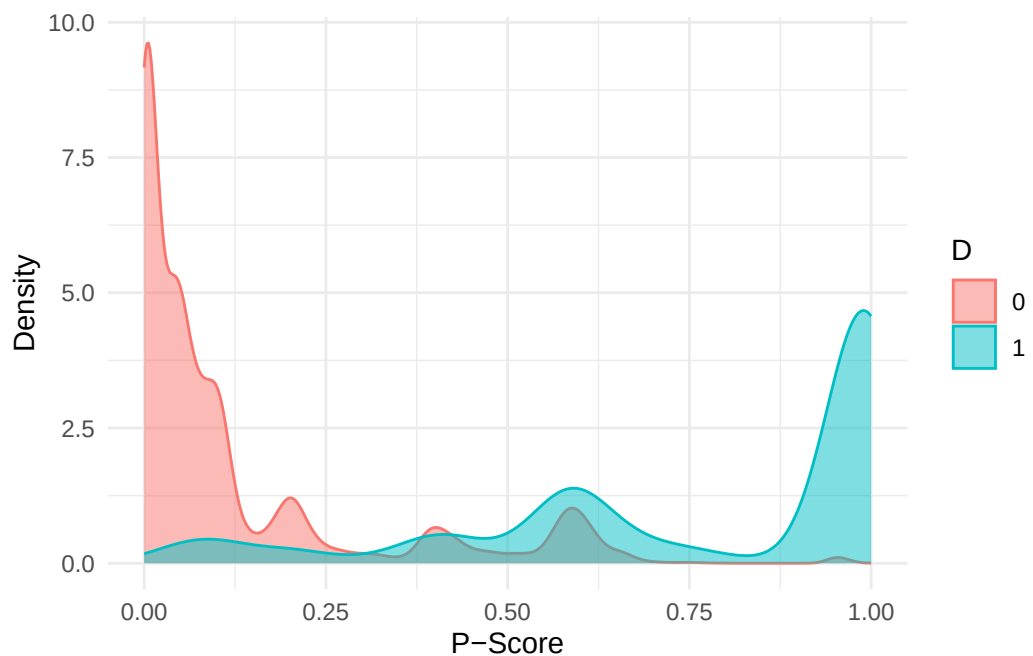
Solution

```
p_score_mod <- glm(
  D ~ Inc + PercentHispanic + PercentBlack
  + per_collegegrads + density
  + factor(State),
  data = data,
  family = binomial(link = "logit")
)
data$p_score <- predict(p_score_mod, type = "response")
```

3.7 In a single well-formatted and compelling visualisation, show the propensity score distributions for the two groups. Based on the plot, which estimand can we identify?

💡 Solution

```
ggplot(data) +  
  aes(x = p_score, fill = factor(D), color = factor(D)) +  
  geom_density(alpha = 0.5) +  
  xlab("P-Score") +  
  ylab("Density") +  
  labs(fill = "D", color = "D") +  
  theme_minimal()
```



The density plot shows that there are treatment units across the full range of the propensity score. However, we observe almost no untreated units with a propensity score between 0.7 and 1, which suggests a lack of overlap. Hence, we can't estimate the ATE or ATT for the full sample. We can still learn something about the causal effect of the treatment but our estimand will be a Conditional Average Treatment Effect (CATE) for the subpopulation of units with a propensity score below 0.7.

3.8 The code below uses the `MatchIt` package to perform nearest neighbour matching without replacement (1:1 matching). Go through the code and documentation to make sure you understand what each line does. Looking at the balance diagnostics, do you think the matching procedure was successful in balancing the covariates?

```
# create an analysis dataset that only includes the covariates used
# for matching and the treatment variable

analysis_data <- data |>
  dplyr::select(
    Cont, D, Inc, PercentHispanic, PercentBlack,
    density, per_collegegrads, p_score
  )

# Nearest neighbour matching (1:1, default)
m_nn <- matchit(
  D ~ Inc + PercentHispanic + PercentBlack +
    density + per_collegegrads,
  data = analysis_data,
  method = "nearest",
  ratio = 1,
  distance = analysis_data$p_score,
  estimand = "ATT"
)

# Check balance
summary(m_nn, un = T)
```

Call:

```
matchit(formula = D ~ Inc + PercentHispanic + PercentBlack +
  density + per_collegegrads, data = analysis_data, method = "nearest",
  distance = analysis_data$p_score, estimand = "ATT", ratio = 1)
```

Summary of Balance for All Data:

	Means Treated	Means Control	Std. Mean Diff.	Var. Ratio
distance	0.7694	0.1288	2.1923	2.5079
Inc	41.1414	37.0118	0.2578	1.1990
PercentHispanic	0.0317	0.0941	-0.7387	0.2485
PercentBlack	0.0525	0.0943	-0.2983	0.5886
density	909.3181	884.1163	0.0092	1.1263

per_collegegrads	19.3481	17.1799	0.1540	1.2263
	eCDF Mean	eCDF Max		
distance	0.4478	0.7562		
Inc	0.0848	0.1452		
PercentHispanic	0.1682	0.2434		
PercentBlack	0.1144	0.1707		
density	0.0222	0.0444		
per_collegegrads	0.0523	0.0800		

Summary of Balance for Matched Data:

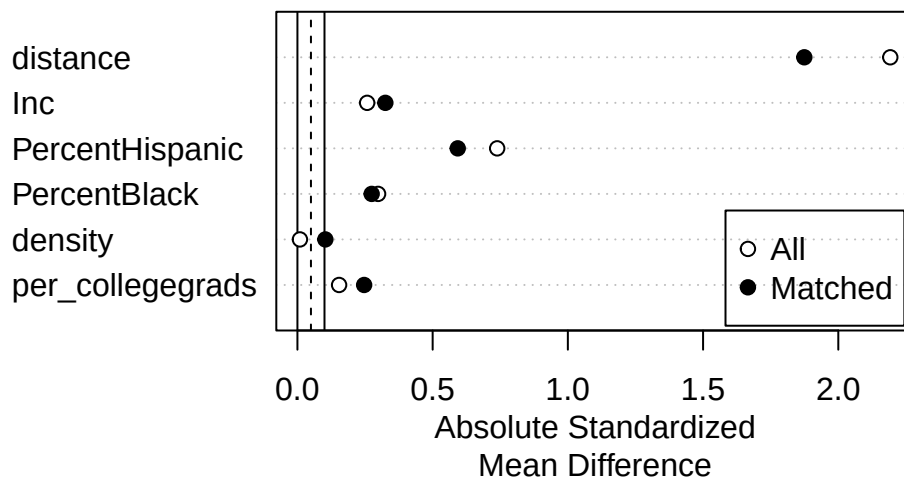
	Means Treated	Means Control	Std. Mean Diff.	Var. Ratio
distance	0.7694	0.2217	1.8743	2.0709
Inc	41.1414	35.9347	0.3250	1.5503
PercentHispanic	0.0317	0.0818	-0.5928	0.2418
PercentBlack	0.0525	0.0910	-0.2744	0.6470
density	909.3181	626.1729	0.1032	1.1644
per_collegegrads	19.3481	15.8763	0.2466	1.4687

	eCDF Mean	eCDF Max	Std. Pair Dist.
distance	0.2929	0.6679	1.8743
Inc	0.0989	0.1634	0.9214
PercentHispanic	0.0975	0.1504	1.0988
PercentBlack	0.1078	0.1627	0.8710
density	0.0496	0.0965	0.4910
per_collegegrads	0.0807	0.1230	0.9376

Sample Sizes:

	Control	Treated
All	11165	6237
Matched	6237	6237
Unmatched	4928	0
Discarded	0	0

```
plot(summary(m_nn, un = T))
```



```
# check specs
print(m_nn)
```

```
A `matchit` object
- method: 1:1 nearest neighbor matching without replacement
- distance: User-defined - number of obs.: 17402 (original), 12474 (matched)
- target estimand: ATT
- covariates: Inc, PercentHispanic, PercentBlack, density, per_collegegrads
```

```
# Extract matched data
matched_data <- match.data(m_nn)

# Estimate the ATT using linear regression on the matched data
lm_robust(Cont ~ D, data = matched_data) |>
  modelsummary(
    stars = T,
    gof_map = c("nobs", "r.squared"),
    output = "markdown"
  )
```

	(1)
(Intercept)	12.811*** (0.891)
D	7.350*** (1.725)
Num.Obs.	12474

	(1)
R2	0.001
	• $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Solution

First, we use `MatchIt::matchit()` to perform nearest neighbour matching. The formula `D ~ Inc + PercentHispanic + PercentBlack + density + per_collegegrads` specifies that `D` is our treatment and that `Inc + PercentHispanic + PercentBlack + density + per_collegegrads` is our conditioning set. We use the propensity scores we estimated earlier as the distance measure for matching. The `ratio` argument controls how many control units should be matched to each treated unit. We specify `estimand = "ATT"` because we are interested in estimating the Average Treatment Effect on the Treated.

Second, we check the balance of covariates before and after matching using `summary(m_nn, un = T)` and `plot(summary(m_nn, un = T))`. The summary provides numerical diagnostics of balance, while the plot visually shows the standardized mean differences for each covariate before and after matching.

Finally, we extract the matched data using `match.data(m_nn)` and estimate the ATT using a linear regression of `Cont` on `D` in the matched sample.

Looking at the balance diagnostics, we see that the standardized mean differences for all covariates are still quite large after matching, indicating that the matching procedure was not very successful in balancing the covariates. For some covariates, the standardized mean differences are even larger after matching (Income, density, and percent college graduates).

- 3.9 Reproduce the steps above, but this time compute the distance matrix using Mahalanobis distance and allow for replacement. Allow up to four control units to be matched to one treated unit. Do we obtain better balance with this approach?

💡 Solution

```
# Mahalanobis distance matching with replacement
m_mahal <- matchit(
  D ~ Inc + PercentHispanic + PercentBlack +
    density + per_collegegrads,
  data = analysis_data,
  method = "nearest",
  ratio = 4,
  distance = "mahalanobis",
  estimand = "ATT",
  replace = T
)

# Check balance
summary(m_mahal, un = T)
```

Call:

```
matchit(formula = D ~ Inc + PercentHispanic + PercentBlack +
  density + per_collegegrads, data = analysis_data, method = "nearest",
  distance = "mahalanobis", estimand = "ATT", replace = T,
  ratio = 4)
```

Summary of Balance for All Data:

	Means Treated	Means Control	Std. Mean Diff.	Var. Ratio
Inc	41.1414	37.0118	0.2578	1.1990
PercentHispanic	0.0317	0.0941	-0.7387	0.2485
PercentBlack	0.0525	0.0943	-0.2983	0.5886
density	909.3181	884.1163	0.0092	1.1263
per_collegegrads	19.3481	17.1799	0.1540	1.2263

	eCDF Mean	eCDF Max
Inc	0.0848	0.1452
PercentHispanic	0.1682	0.2434
PercentBlack	0.1144	0.1707
density	0.0222	0.0444
per_collegegrads	0.0523	0.0800

Summary of Balance for Matched Data:

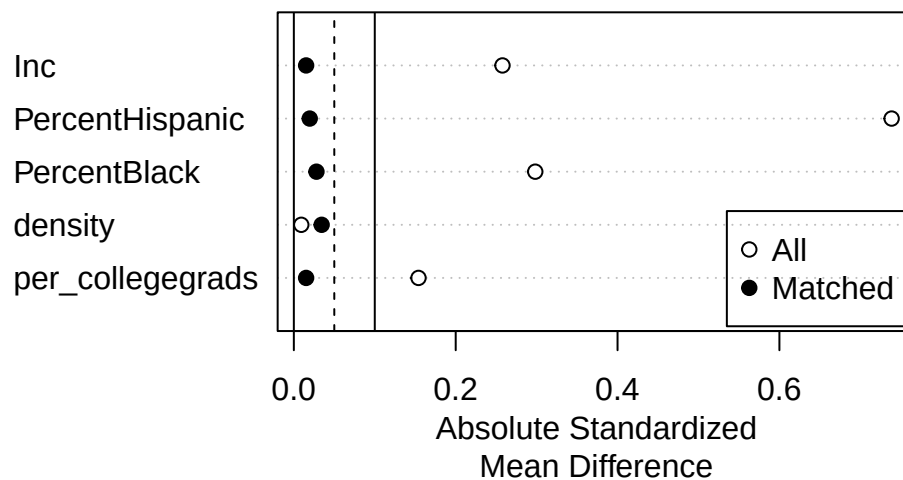
	Means Treated	Means Control	Std. Mean Diff.	Var. Ratio
Inc	41.1414	40.8982	0.0152	1.0429
PercentHispanic	0.0317	0.0334	-0.0197	0.9645

PercentBlack	0.0525	0.0486	0.0279	1.0756
density	909.3181	814.9917	0.0344	1.1574
per_collegegrads	19.3481	19.1344	0.0152	1.0635
	eCDF Mean	eCDF Max	Std. Pair	Dist.
Inc	0.0042	0.0127		0.0951
PercentHispanic	0.0156	0.0352		0.1294
PercentBlack	0.0173	0.0560		0.0831
density	0.0452	0.0826		0.0811
per_collegegrads	0.0031	0.0109		0.0907

Sample Sizes:

	Control	Treated
All	11165.	6237
Matched (ESS)	4371.17	6237
Matched	7528.	6237
Unmatched	3637.	0
Discarded	0.	0

```
plot(summary(m_mahal, un = T))
```



With the modified matching procedure, we do obtain much better balance. The standardized mean differences for all covariates are now much closer to zero, indicating that the distribution of covariates is more similar between the treated and control groups after matching. This suggests that the Mahalanobis distance matching with replacement has been more effective in creating a balanced sample compared to nearest neighbour matching without replacement.

3.10 Estimate the ATT using linear regression on the matched data obtained from Mahalanobis distance matching with replacement. How does this estimate compare to the one obtained with nearest neighbour matching without replacement?

 Solution

```
# Extract matched data
matched_data_mahal <- match.data(m_mahal)
# Estimate the ATT using linear regression on the matched data
lm_robust(Cont ~ D, data = matched_data_mahal,
  weights = matched_data_mahal$weights) |>
  modelsummary(
    stars = T,
    gof_map = c("nobs", "r.squared"),
    output = "markdown"
  )
```

	(1)
(Intercept)	22.683*** (1.702)
D	-2.521 (2.254)
Num.Obs.	13765
R2	0.000

The ATT estimate using Mahalanobis distance matching with replacement is -2.521, which is not statistically significant at conventional levels. This estimate is smaller in magnitude compared to the nearest neighbour matching estimate, likely due to the improved balance achieved with Mahalanobis distance matching.

- 3.11 Estimate the ATT using the Lin-Estimator. You may use the `lm_lin()` function from the `estimatr` package or write your own code to implement the Lin-Estimator. Use the original unmatched data and control for the same covariates used for matching. How does the estimate differ from the one obtained with matching? What could explain the difference in estimates?

💡 Solution

```
lm_lin(Cont ~ D,
  covariates = ~ Inc + PercentHispanic + PercentBlack +
    density + per_collegegrads + factor(State),
  data = data) |>
  modelsummary(
    stars = T,
    coef_map = c("D" = "ATT"),
    gof_map = c("nobs", "r.squared"),
    output = "markdown"
  )
```

	(1)
ATT	-4.033+ (2.217)
Num.Obs.	17402
R2	0.305

The Lin estimator adjusts for covariates by centering them and including treatment-by-covariate interactions. This approach is more flexible than standard OLS as it allows for heterogeneous treatment effects across covariate values. The difference between the Lin estimate and the matching estimates could be explained by the trimming of the sample that occurs with matching (since units without overlap are dropped). This trimming also means that the two estimators may not be estimating the same estimand, as the matching estimator is conditioned on overlap. Additionally, matching is a non-parametric method that is agnostic to functional form, while the Lin estimator relies on a parametric specification, albeit a more flexible one than standard OLS due to the inclusion of interaction terms.

i Feedback

Feedback to the problem set will be provided here.